Predictive Modeling of Heart Disease Using Binary Logistic Regression

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***Abstract*—This study focuses on the application of Binary Logistic Regression to predict heart disease using a clinical dataset. The goal of the research was to understand how model settings specifically regularization, feature selection, and decision thresholds impact prediction accuracy. Through four experiments, I found that using a complete set of 11 clinical features and a balanced regularization strength (C=1.0) produced the most reliable results. The final model achieved a mean accuracy of 85.00% through 5-fold cross-validation, demonstrating that logistic regression is an effective tool for medical screening.**

Keywords*—Binary Logistic Regression, Sigmoid Function, Heart Disease Prediction, Machine Learning in Healthcare, Regularization (C-parameter), K-Fold Cross-Validation, Performance Metrics.

I. Introduction

Logistic regression is a fundamental classification algorithm used when the target variable is binary, such as "Disease" vs. "No Disease". While Linear Regression predicts continuous values, it is unsuitable for classification because it can produce values outside the 0 to 1 range.

Logistic regression solves this by using the Sigmoid Function:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

This function squashes the output (log-odds) into a probability between 0 and 1. If the probability is \geq 0.5, the model classifies the result as 1 (Heart Disease); otherwise, it is 0.

Fig. 1. Sigmoid Function Formula

II. Methodology

A. Dataset Description

The dataset used in this lab contains clinical observations with 11 independent variables, including age, sex, cholesterol levels, and heart rate (thalach). The target variable is binary (Presence/Absence of disease).

B. Data Preparation

Before training, I cleaned the dataset by removing duplicates and handling missing values to ensure the results weren't skewed. I then split the data using an 80/20 ratio, where 80% was used for training the models and 20% was reserved strictly for evaluating performance.

- **Cleaning:** Addressed missing values and removed duplicates.
- **Partitioning:** The data was split into 80% training and 20% testing sets.

III. Experimental Results and Discussion

A. Experiment 1: Hyperparameter Tuning (C Parameter)

I tested the C parameter to see how regularization affects the model. A small C means stronger regularization (simpler model), while a large C allows for more flexibility.

Model Performance Comparison

Experiment	C Value	Accuracy	Precision
Strong Regularization	0.001	0.6441	0.0000
Baseline	1.0	0.7966	0.7368
Weak Regularization	1000.0	0.7797	0.7222

Fig. 2. Hyperparameter Tuning Experiment Results

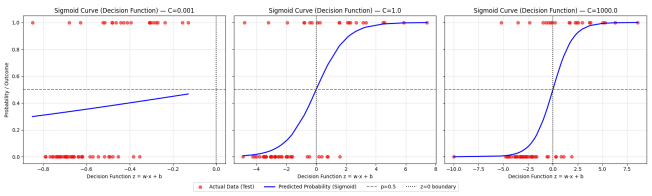

Fig. 3. Sigmoid Curves for Varying C Values (0.001, 1.0, 1000.0)

As shown in Figure 3, a smaller C produces a flatter sigmoid curve because the weights are smaller. The baseline (C=1.0) performed best. Strong regularization (C=0.001) actually made the model too simple, resulting in zero recall because it couldn't identify any positive cases.

B. Experiment 2: Feature Engineering and Selection

I tested how adding or removing variables changed the predictions.

Model Version	Features Used	Accuracy	Precision
Model A	thalach	0.5424	0.2857
Model B	age, chol, thalach	0.5424	0.3125
Model C	**All 11 Features**	**0.8136**	**0.7500**

Fig. 4. Feature Selection Results

Model C had the strongest influence on prediction. This proves that one or two variables (like heart rate or age) aren't enough to predict heart disease accurately; you need the full clinical context.

C. *Experiment 3: Decision Threshold Adjustment*

I changed the default 0.5 threshold to 0.7 to see the trade-off between False Positives (FP) and False Negatives (FN).

Threshold	Accuracy	Precision	Recall
0.7	0.8305	0.9231	0.5714

Fig. 5. Threshold Adjustment (Threshold = 0.7)

When the threshold increases, Precision goes up, but Recall drops, leading to more False Negatives. In the medical domain, False Negatives are much more dangerous because a sick person might be told they are healthy and miss out on treatment.

D. *Experiment 4: K-Fold Cross-Validation*

To make sure the model was stable and didn't just get lucky with the data split, I ran a 5-fold cross-validation.

Fig. 6. Model Performance Stability Across Folds

The accuracy was consistent across all folds, ranging from 81% to 89%. The mean accuracy was 0.8500, which gives me confidence that the model generalizes well to different patients.

IV. Performance Evaluation

My best-performing model (Model C with C=1.0) showed a strong balance across all metrics.

Metric	Mean	Std Dev
Accuracy	0.8499	0.0321
Precision	0.8486	0.1046
Recall	0.7346	0.1115
F1 Score	0.7782	0.0549

Fig. 7. K-Fold Summary (Mean/Std Dev)

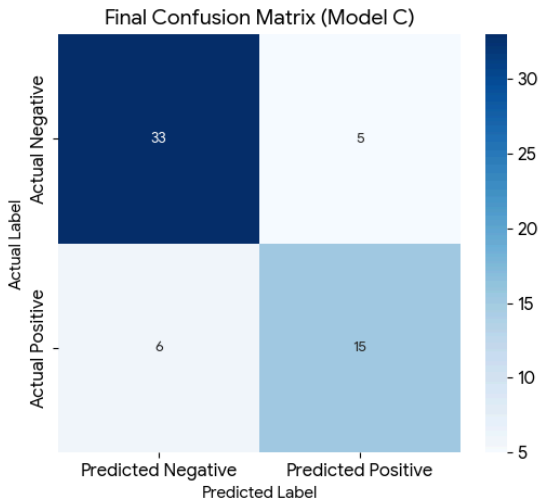

Fig. 8. Final Confusion Matrix (Model C)

V. Conclusion

The most important predictor in this study was the combination of all 11 features, as seen in Model C. The best model settings were a C value of 1.0 and maintaining a 0.5 threshold to avoid the high False Negative rate found at 0.7. Practically, this model shows that while individual tests are helpful, digital tools using logistic regression can effectively assist doctors in screening for heart disease by looking at the "big picture" of a patient's health data.